Patrick Ferguson

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Research Interests

Econometrics; partial identification; statistical decision theory under ambiguity; survey design

Education

Cornell University — Expected 2027
PhD Economics
Committee: Jörg Stoye (Chair), Francesca Molinari, José Luis Montiel Olea

University College Dublin — 2020
MSc Quantitative Economics

Trinity College Dublin — 2019
BA Economics and Psychology

Job Market Paper

Paying to Sharpen Identification: Optimal Probing of Coarse Data

Summary. A researcher observes coarse data and must select a limited number of observations to probe under a budget constraint. I show that for certain target parameters, we can quantify the contribution of each coarse observation to identification uncertainty. This implies an optimal probing rule which probes observations in decreasing order of expected diameter reduction per unit cost, until the budget is exhausted. The rule is optimal both for scalar expectations and, under Wasserstein-1 distance, for univariate distributions. I demonstrate the rule in two survey-design problems.

Working Papers

Statistical Treatment Choice with Untested Doses

Asymptotic Minimax Optimality of Plug-In Rules
with José Luis Montiel Olea and Jörg Stoye

Research & Teaching Experience

Research Assistantships

Francesca Molinari and Levon Barseghyan — Summer 2025
Stephen Coate — Summer 2023

Teaching Assistantships

Mathematical Economics (PhD) — Fall 2022–2025
Economics of Wages and Employment (UG) — Spring 2025
Microeconomic Theory II (PhD) — Spring 2024
Intermediate Microeconomics (UG) — Spring 2023

Awards

The Louis Walinsky Fund in Economics in Honor of Professor Herbert Joseph Davenport Outstanding Teaching Award — 2023

Sage Fellow, Cornell University — 2021–present

Denis Conniffe Medal, UCD — highest overall score in the MSc programme — 2020

Additional Information

Citizenship: Ireland

Software: LaTeX, R (proficient); Julia, Stata, MATLAB, Python (some experience).

References

Jörg Stoye (Chair)
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